3.1. Biochemical molecules

At the end of this section, the student will be able to:

- **identify inorganic and organic biochemical molecules,**
- **discuss different inorganic ions and their contribution to the cell**
- **explain the properties of water and its importance for life**

3.1.1. Inorganic molecules: water

At the end of this section, the student will be able to:

- **mention properties of water,**
- **discuss the importance of water for the living organisms.**

3.1.2. Inorganic ions

At the end of this section, the student will be able to:

- **mention important inorganic ions that are necessary for life,**
- **mention diets that contain inorganic ions,**
- **discuss the effect of deficiency of nutrient elements in our body,**
- **explain the importance of each nutrient element to the human body.**

3.1.3. Organic molecules

At the end of this section, the student will be able to:

- **discuss the role of biological molecules in the cell structure,**
- **Classify organic molecules based on their constituent elements and the monomers from which they are constructed.**
- **elaborate the functions of carbohydrates, proteins, lipids, and Nucleic acids to the body,**
- **conduct experiments to identify nutrients in different foodstuffs,**
- **appreciate why Ethiopians use malting seeds to make local drinks (Tella, Areke), and**

- **appreciate the significance of local drinks prepared from malting seeds in Ethiopia.**

Carbohydrates

At the end of this section, the student will be able to:

- **name the different types of carbohydrates,**
- **discuss the importance of monosaccharides, and disaccharides,**
- **explain which food items are sources of carbohydrates, and**
- **tell the types and importance of polysaccharides**

Lipids

At the end of this section, the student will be able to:

- **discuss what lipids are**
- **tell the importance of lipids for our cells,**
- **describe the structure of glycerol and fatty acids, and**
- **explain the importance of phospholipids in the cell membrane.**

Proteins

At the end of this section, the student will be able to:

- **mention the importance of proteins to cells and the body of living things,**
- **elaborate formation of proteins by amino acids,**
- **differentiate proteins according to their structure,**
- **define what are amino acids, and**
- **group proteins according to their function.**

Nucleic acids

At the end of this section, the student will be able to:

- **describe the structures of nucleotides and nucleic acids,**
- **differentiate between DNA and RNA, and**
- **explain the importance of DNA and RNA.**